



KunHo Heo

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Research Interests

Scene Understanding

- 3D Scene Graph, 3D Affordance Grounding

Character Animation

- Text-driven Motion Generation, Human-Object Interaction

Perception

- Few-Shot Object Detection, 3D Anomaly Detection

Education

Kyung Hee University

Mar. 2025 – Feb. 2027 (Expected)

M.S. in Software Convergence

- GPA: 4.46 / 4.5

Kyung Hee University

Mar. 2021 – Feb. 2025

B.S. in Software Convergence

- GPA: 4.04 / 4.5

Research Experience

Visual Science Lab [🔗](#), Kyung Hee University

Mar. 2025 – Present

M.S. Student (**Lab Manager**)

- Advisor: Prof. MyeongAh Cho

Visual Science Lab [🔗](#), Kyung Hee University

Sep. 2024 – Feb. 2025

Undergraduate Research Intern

- Advisor: Prof. MyeongAh Cho

*IIIXR Lab [🔗](#), Kyung Hee University **relocated to Korea University*

Jan. 2023 – Jun. 2024

Undergraduate Research Intern

- Advisor: Prof. HyeongYeop Kang

Publications

International Conference

Towards Generalizable 3D Anomaly Detection via Relational Inconsistency Modeling

KunHo Heo^{*}, SuYeon Kim^{*}, Hayoung Lee, Chanse Oh, MyeongAh Cho **Equal Contribution*

Under Review

Rethinking Prototype-based Similarity Learning for Few-Shot Object Detection

[Paper](#) [🔗](#)

KunHo Heo^{*}, Seungjae Kim^{*}, Wongyu Lee^{*}, SuYeon Kim, MyeongAh Cho **Equal Contribution*

European Conference on Computer Vision (**ECCV 2026**)

ParTY: Part-Guidance for Expressive Text-to-Motion Synthesis

Paper [↗](#)

KunHo Heo, SuYeon Kim, Yonghyun Gwon, Youngbin Kim, MyeongAh Cho

IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2026**)

Object-Centric Representation Learning for Enhanced 3D Semantic Scene Graph Prediction

Paper [↗](#)

KunHo Heo^{*}, GiHyun Kim^{*}, SuYeon Kim, MyeongAh Cho ^{*}*Equal Contribution*

Conference on Neural Information Processing Systems (**NeurIPS 2025**)

International Journal

Align What Matters: Selective Source-Oriented Audio-Visual Alignment for Sound Source Localization

Wongyu Lee, Youngbin Kim, **KunHo Heo**, MyeongAh Cho

Under Review (*IEEE Transactions on Multimedia*)

Revisiting Object-Centric Learning for Long-Tailed 3D Semantic Scene Graph Prediction

GiHyun Kim, **KunHo Heo**, SuYeon Kim, MyeongAh Cho

Under Review (*IEEE Transactions on Image Processing*)

Domestic Conference

Word-Motion Cross Attention for Fine-Grained Text Guided Human-Human Interaction Motion Generation

Yonghyun Gwon, **KunHo Heo**, Youngkwan Cho, SuYeon Kim, Hayoung Lee, MyeongAh Cho

Korea Software Congress (**KSC**), 2025

Distance-based Bi-directional Edge Update Graph Neural Network for 3D Scene Graph Generation

SuYeon Kim, GiHyun Kim, **KunHo Heo**, MyeongAh Cho

Korea Computer Congress (**KCC**), 2025

Enhancing Text-driven Human Motion Generation via Part Transfer and Sequential Attention Learning

KunHo Heo, Seung Joon Lee, Hayoung Lee, MyeongAh Cho

Korea Software Congress (**KSC**), 2024

Body-Part Embedding for Text-driven Human Motion Generation

KunHo Heo, HyeongYeop Kang

Korea Computer Congress (**KCC**), 2024

Personality labeling methods with motion parameters

KunHo Heo, HyeongYeop Kang

Korea Software Congress (**KSC**), 2023

Projects

Development of AGI platform for multisensory social robot

Institute for Information communication
Technology Planning and Evaluation (IITP)
Apr. 2025 - Dec. 2025

- **Project Manager**
- Developed the **Vision system** for multisensory perception
- Researched **Few-Shot Object Detection** to enhance adaptability to dynamic environments under limited data

Development of AI technology capable of robustly recognizing abnormal and dangerous situations and behaviors during night and bad weather conditions

Korea Planning & Evaluation Institute
of Industrial Technology (KEIT)
Jul. 2024 - Mar. 2028

- **Image restoration** and **Data pre-processing** under adverse visual conditions

Anomaly pre-detection and response for intelligent surveillance system: high-level interpretation and response based on the visual-language fusion

National Research Foundation
of Korea (NRF)
Sep. 2024 - Aug. 2027

- **Vision-language model integration** for surveillance anomaly interpretation and response

Patents

Domestic Patent

Method and Apparatus for Generating Whole-Body Motion by Synthesizing Part-wise Motions Dec. 2025

- *KR-Application No. 10-2025-0200238*

Method and Apparatus for Generating 3D Scene Graphs Dec. 2025

- *KR-Application No. 10-2025-0188361*

Software Copyright

Improved Sound Source Localization Model via Audio Signal Enhancement Sep. 2025

- *KR-Registration No. C-2025-035472*

Honors & Awards

Outstanding Researcher Scholarship 2025, 2026

- *Kyung Hee University*

Academic Scholarship 2021 – 2024

- *Kyung Hee University*

Grand Prize Dec. 2024

- *Undergraduate Paper Competition, Korea Software Congress (KSC)*

Grand Prize (Overall 1st Prize across all categories) Nov. 2024

- *SW Festival, Kyung Hee University*

1st Prize in Immersive Media Category Nov. 2024

- *SW Festival, Kyung Hee University*

National Research Foundation (NRF) President's Award Nov. 2024

- *Immersive Media Competition, Convergence and Open Sharing System (COSS)*

Grand Prize May. 2024

- *AI&SW Hackathon, Kyung Hee University*

1st Prize in Immersive Media Category Nov. 2023

- *SW Festival, Kyung Hee University*

Teaching Experience

Teaching Assistant Spring 2025, Spring 2026

- Artificial Intelligence and Game Programming

Teaching Assistant Fall 2025

- Game Engineering
- Open Source Software Development Methods and Tools
- Design Thinking